

Initial Funding Request Form

Approved by EPAC Chairmen external to meeting on 06/11/2021

[Link to 06/23/2021 EPAC Meeting Minutes](#)

Date Prepared: 06/10/2021	Project Title: Hyde Park BESS – Battery Energy Storage System Solution
Company/ies: Eversource EMA	Project Number: 21334
Organization: System Planning	Class(es) of Plant: D SS
Project Initiator: Ling Yang	Project Category: Stations - Other
Project Manager: Anuj Mathur	Project Type: Specific
Project Sponsor: Jennifer A Schilling	Project Purpose: Evaluate a new BESS across multi-voltage-level system and alleviate area loading area capacity
Estimated in service date: Q3 2025	If Transmission Project (check all that apply): PTF ☐ Non-PTF ☐
Authorization Type: Initial Funding	
Total Request: \$200K	

Project Need Statement:

An initial funding request of \$200,000 is requested to fund initial engineering studies for installation of a 15 MW/20 MWh utility-scale battery storage system (BESS) along Cummins Highway near Hyde Park Substation #496 in Hyde Park, MA. See attached Figure 1 for the geographic layout of the Hyde Park area and proposed BESS location and Figure 2 for details.

The proposed project is to test a new concept of installing a BESS connected to multiple voltage levels, moving energy between Hyde Park #496 and Baker St#110 substations with different voltage levels and test the charge and discharge of the battery at the same time. This will also help relieve heavy loading condition of Hyde Park Substation #496. Hyde Park substation #496 is heavily loaded above 95% of the substation load carrying capability (LCC). A near-term load transfer and a mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192 - Hyde Park-Dorchester Area Supply Long-Term Plan) will be in service.

The proposed BESS installation will not only test new BESS applications but can also serve as the mid-term solution to bring the Hyde Park substation loading below 95% of the substation firm LCC.

Project Objectives:

The objective of the project is to test new application/usage of the battery storage system and increase the Hyde Park area capacity. The project is expected to demonstrate new BESS design/application ideas. New applications include testing the connected BESS system to

various system voltage levels, (14kV and 25kV distribution systems); moving energy from one (1) substation to the other when two (2) systems cannot be directly connected; and testing partial discharging of the BESS while the remaining portion is charging with grid.

The battery storage projects are expected to be authorized under the MA Department of Utilities (MA DPU) "Grid Modernization - 2.0" docket MDPU 15-122/15-123. See attached Figure 3 for a simplified one-line diagram. See attached Figure 4 for the area feeder diagram and potential BESS site.

The benefits of the proposed design are as follows:

- Pilot project to test connecting BESS to multi-level voltage systems
- Test moving energy through BESS between substations from one (1) to the other when two (2) systems cannot be directly connected
- Test partial BESS charging with grid while the remaining portion is discharging, which could potentially reduce the MWh needed compared to a system that cannot charge/discharge simultaneously.
- Keep transformer cooler for pre-contingency event
- Avoid load profile being flat at heavily loaded substation
- Bring additional capacity to Hyde Park #496; Reduce the need to perform break before make emergency load switching and serve as the mid-term solution to prevent substation from overload
- Provide additional capacity to Baker St#110 during equipment maintenance in the near-term and load relief in the long-term
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Alternative:

- One alternative to be evaluated as part of this project is to construct new distribution lines from Needham station to bring an additional 15MVA of capacity to Hyde Park. These options will be presented to SDC to determine which solution should be pursued.

Funding Request Explanation (total request, amount per task, deliverables):

A detailed project scope including quantities, materials, manufacturer selection, has not been conducted yet, and detailed feedback from all affected disciplines (Station Engineering, Control and Protection, Communication Engineering, Environmental, etc.) has not yet been attained as part of this project. This \$200K initial funding request will be used to develop preliminary engineering, including the following:

Station Engineering - \$45K

Provide station design scope and cost estimates for potential solutions

System Planning – \$30K

Perform system analysis and study to evaluate the system impacts

Distribution Engineering--\$45K

Evaluate street work feasibility, scopes and provide cost estimates for solutions.

Evaluate street work feasibility, scopes and cost estimates to bring an additional 15MVA capacity to Hyde Park area by constructing new distribution lines from Needham Station, and determine when distribution work becomes cost prohibited.

Environmental and Siting - \$25K

Identify environmental and siting requirements for potential solutions

Project Management - \$20K

Collaborate engineering team works

Control and Protection - \$20K

Identify control and protection upgrade requirements for potential designs

Communication Engineering – \$15K

Identify communication scope/cost for potential designs

Preliminary Schedule:

Milestone/Phase Name	Estimated Date
Initial Funding Request – Approved	Q2 2021
Preliminary engineering analysis	Q3 2021
Develop Scope of Work – Draft Solution Selection Form (SSF)	Q3 2021
Solution Design Committee (SDC) Review of Preferred Solution	Q3 2021
Preliminary Engineering Completion	Q1 2022
Full Funding request at EPAC	Q3 2022
Construction Start	Q4 2023
In Service Date	Q3 2025

Figures

Figure 1 - Geographic Layout of the Hyde Park Area

Figure 2 - Geographic Layout of the Hyde Park Area (Detail)

Figure 3 - Simplified One-Line Diagram

Figure 4 - Area Feeder Diagram and potential BESS site

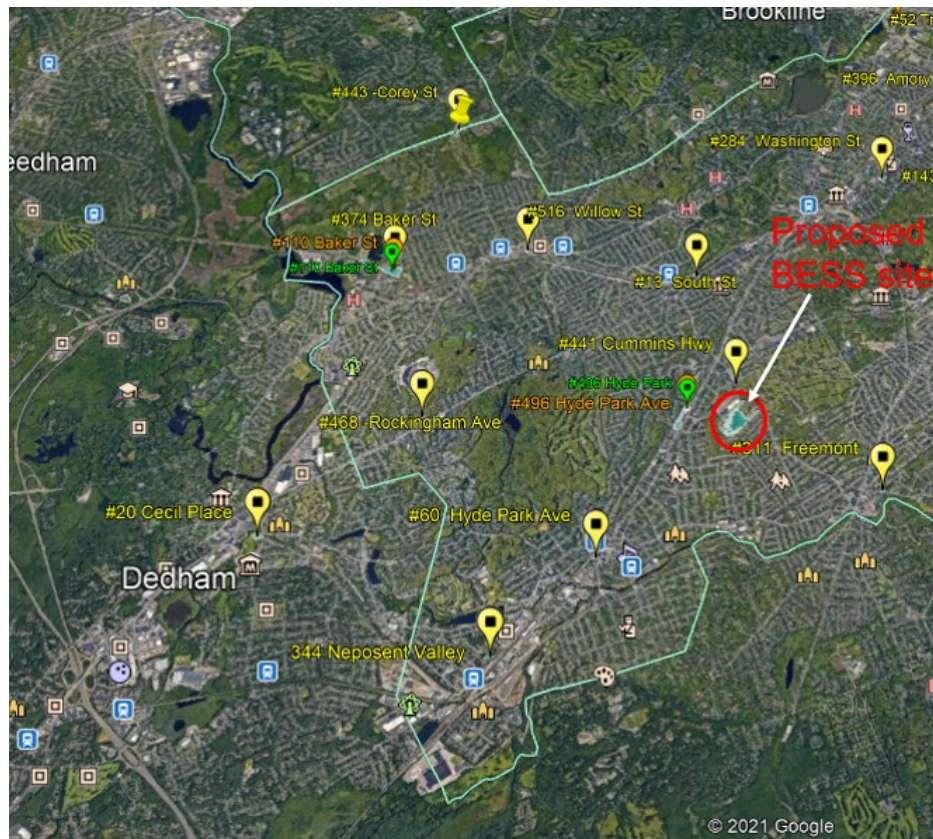


Figure 1 - Geographic layout in the vicinity of Hyde Park and Proposed BESS site

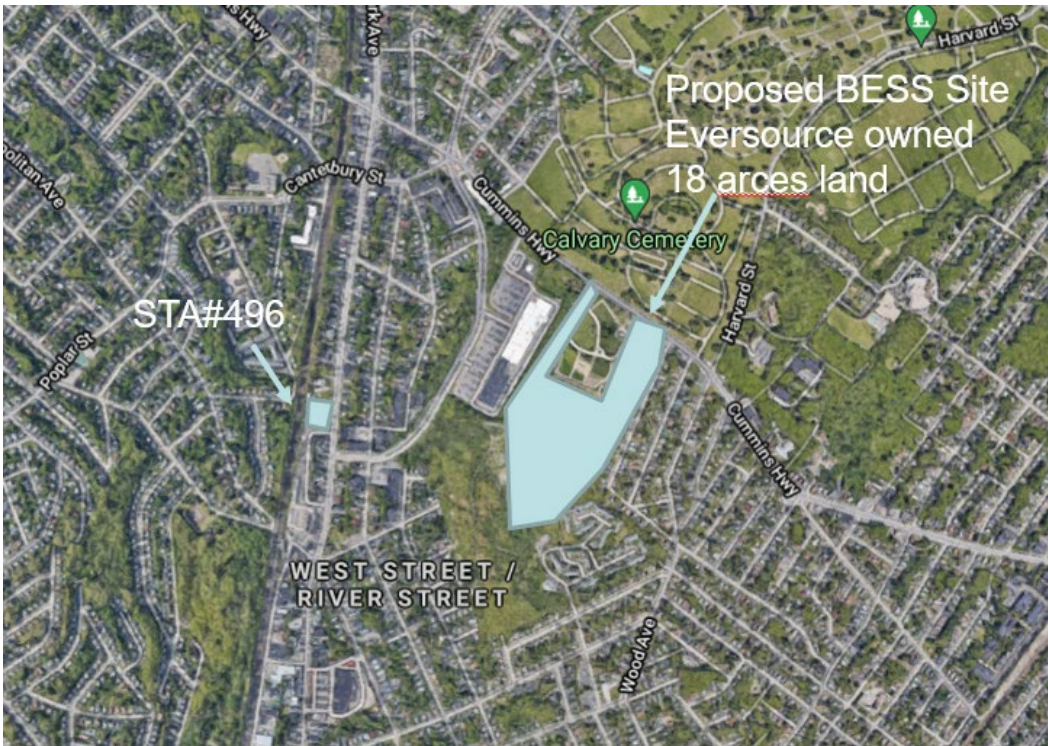


Figure 2 - Geographic layout in the vicinity of Hyde Park and Proposed BESS site (zoom in)

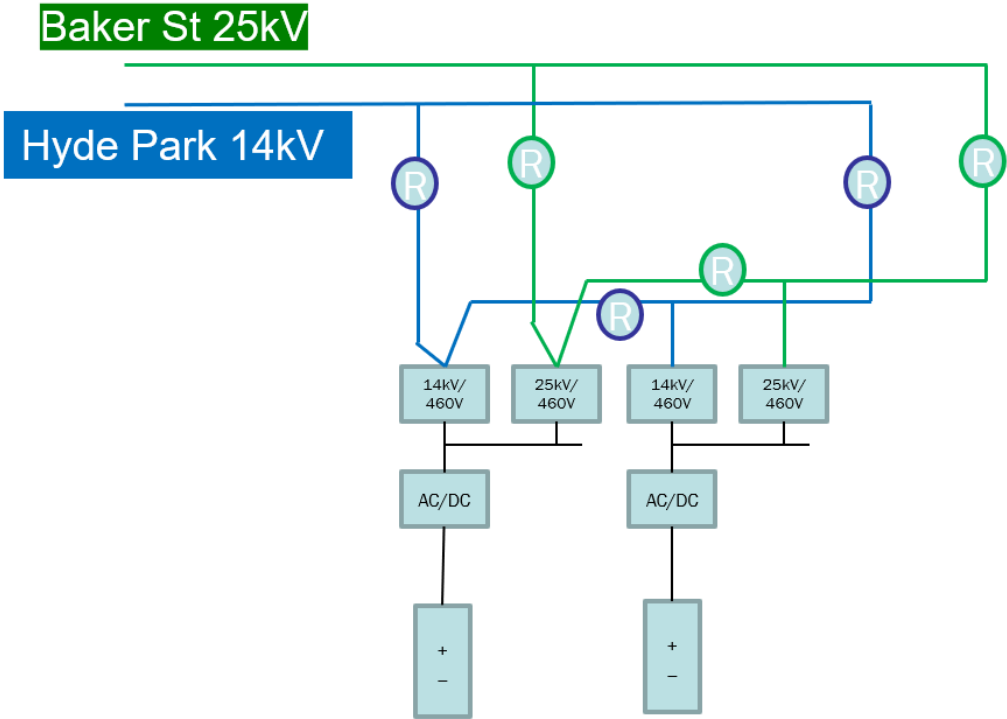


Figure 3 - Simplified One-Line Diagram

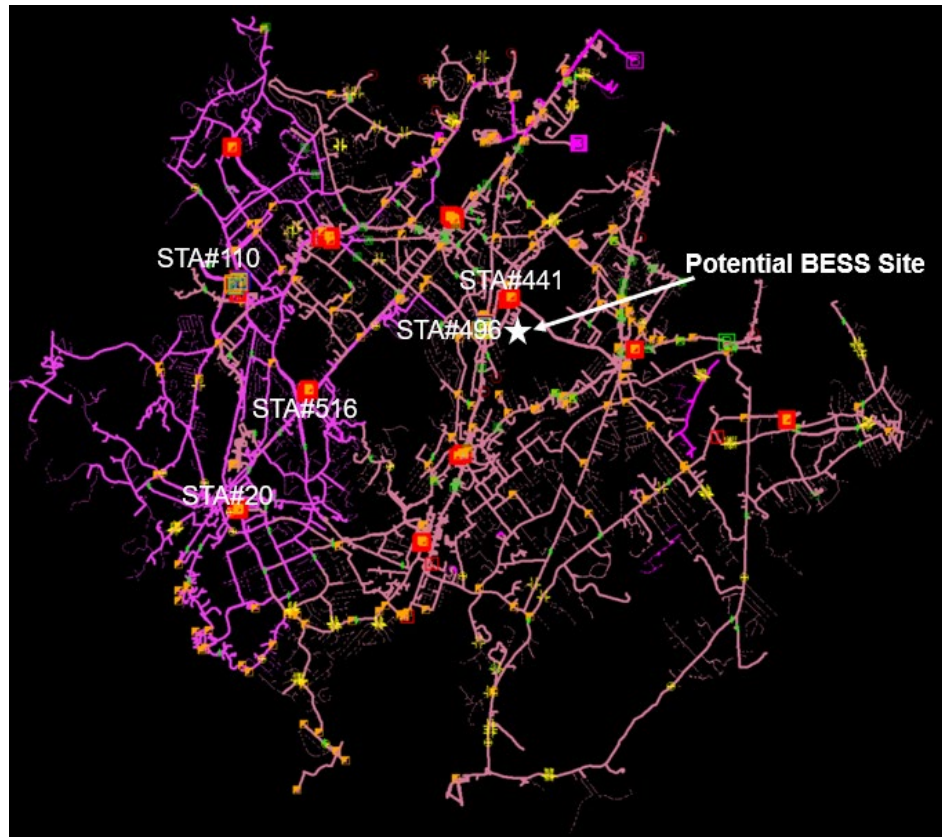


Figure 4 - Area Feeder Diagram and potential BESS site